

similar to Symica from the technology point of view.

Why Symica

The main advantage of Symica over its full-featured competitors would be its pricing factor. While the cost of buying a custom IC design suite can be pretty high, the Symica FE is available for free. For additional functionalities, such as support for mixed signal features, unlimited number of devices, threads and Verilog-A model parameters, full version of Symica can be purchased at an affordable and flexible pricing scheme.

Though Qucs and Ngspice are open source projects, they are not straightforward for compiling and simulating the Verilog-A models. Regardless of the zero or low pricing, Symica offers the user with easy user interface, high performance, high accuracy, advanced features and industry standards comparable to other top-notch tools.

Compatibility with industry standards

Symica is compatible with the industry-standard formats (HSPICE and Spectre) for netlists, models and analysis features. It allows for the import of SKILL format files (file extension *.il) of Cadence Design System, Inc. Accuracy for critical analogue circuits is achieved using a set of numerical methods and models that produce results similar to those of industry-proven standards. The software is available for both Linux and Windows platforms.

Technical support

An updated version of the software is published every few months. Visit the company's website www.symica.com to request a trial version or order a full package of this software. Technical support can be obtained from the website for licenced users. You can also send an e-mail to support@symica.com.

Want to help Symica beta test. You can be a beta tester by contacting them via e-mail. Symica services are also available on social networking services such as Twitter (www.twitter.com/

System Requirements

Operating system (OS)	Linux: 32 bit • OpenSUSE 12.x • Ubuntu 11.10 • Debian 6.x • CentOS/RHEL 4.x, 5.x, 6.x • Fedora 16 64 bit (with 32-bit system libraries installed additionally) • CentOS/RHEL 5.x, 6.x • Fedora 14 Windows: Windows XP, Windows 2K3, Windows Vista and Windows 7
Disk space	120MB (excluding the resources required to run simulation and save the results)
Random access memory (RAM)	At least 1GB to simulate circuits with up to 50,000 elements At least 4GB to simulate circuits with up to 500,000 elements

Installation

Linux	32-bit OS: Open the distribution package and run the file <i>symica_free_stable_x86_2.2.0.0304.run</i> with regular user privilege. In the window that appears, follow instructions for installation. Now set up the path to the executables to the environment variable PATH: <pre>.bashrc: export PATH=\$PATH:<instdir>/bin .cshrc: setenv PATH \$PATH:<instdir>/bin</pre> Run the integrated environment: <pre>\$ symica &</pre> 64-bit OS: Apart from the procedure given above for 32-bit OS, the user should install additional x32 system libraries: <pre>\$ sudo yum install--setopt=protected_multilib=false glib2.i686 glibc.i686 zlib.i686 libXext.i686 libgcc.i686 freetype.i686 libSM.i686 libXrender.i686 fontconfig.i686 libstdc++.i686 gcc-gcc-c++ make glibc-devel.i686</pre> Also ensure that glib2 version is 2.32.1-4 or higher: <pre>\$ yum info glib2</pre>
Windows	Right click on the setup file <i>Symica 2.3.0.1002 Free.exe</i> and choose the option Run As Administrator. In the wizard window that appears follow the instructions for installation.

Note: Installation files for both Linux and Windows platforms are available in the DVD that accompanies this issue of Electronics For You Plus.

symica) and LinkedIn (www.linkedin.com/company/symica). Users can register on the company's website with their e-mail IDs to receive monthly SYMICA custom IC design toolkit newsletter in their inbox.

Cadre Design Systems is the sales partner of Symica in India. You can visit their website www.cadredesign.co.in or send an e-mail to sales@cadredesign.in for support.

Licence

The software provided with the DVD is Symica Free Edition (Symica FE). This is licenced for personal, noncommercial

and nonprofit use, and is intended for research and educational purpose.

Free edition can simulate only up to 300 devices, 10 Verilog-A model parameters and does not offer support for mixed signal features. But this edition does not require any registration and can even be downloaded directly from their site.

Academic licence is available for universities and other educational institutions. Basic, AMS or full licence should be obtained for other commercial uses. A 30-day trial license is also available. ●

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